Practical 6

Aim: Write shell script for the following -

Theory:

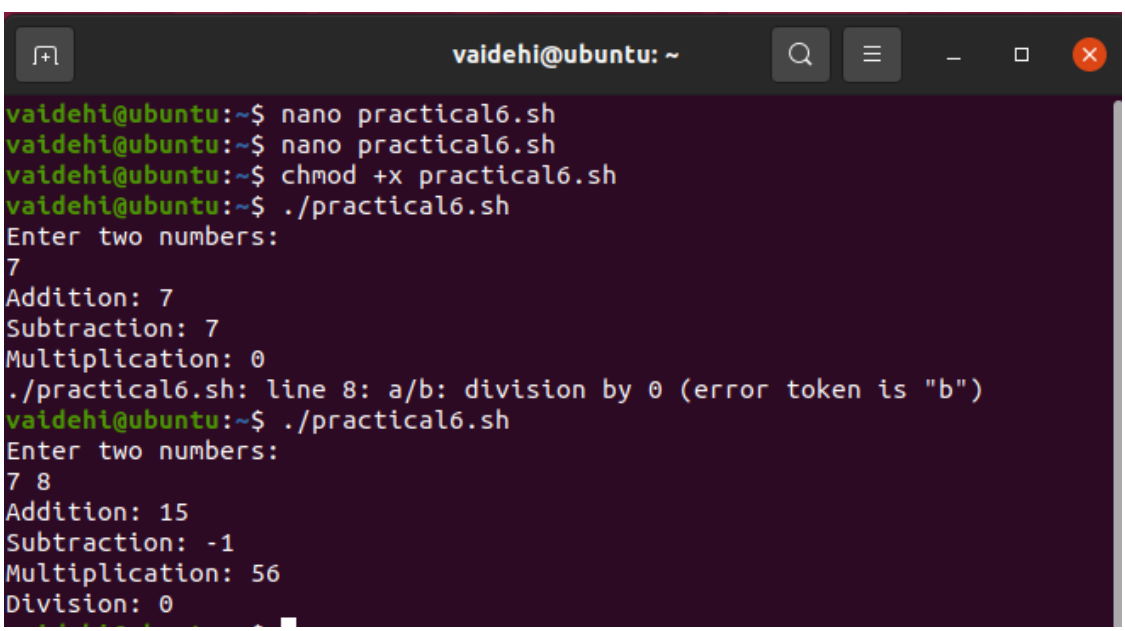
Explain what shell script is in the operating system and its use.

A shell script is a file containing a sequence of commands written for the Linux/Unix shell to execute. It automates repetitive tasks, reduces manual effort, and helps in system administration. Shell scripts are widely used for file handling, system monitoring, backups, and performing calculations, making them an essential tool in operating systems.

- Write a shell script to perform arithmetic operations.

```bash
echo "Enter two numbers:"
read a b

echo "Addition: $((a+b))"
echo "Subtraction: $((a-b))"
echo "Multiplication: $((a*b))"
echo "Division: $((a/b))"
```



- Write a shell script to calculate Simple Interest and Compound Interest calculation.

```bash
echo "Enter Principal, Rate, Time:"
read p r t

si=$((p*r*t/100))
ci=$(echo "$p*(1+$r/100)^$t" | bc)

echo "Simple Interest: $si"
echo "Compound Interest: $ci"
```

```
vaidehi@ubuntu:~$ nano practical6.sh


Use "fg" to return to nano.

[2]+  Stopped                 nano practical6.sh
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$
vaidehi@ubuntu:~$ ./practical6.sh
Enter Principal, Rate, Time:
100000 2 2
Simple Interest: 4000
Compound Interest: 100000
vaidehi@ubuntu:~$
```

- Write a shell script to determine if a given year is a leap year or Not.

```bash
echo "Enter year:"
read year

if (( (year%4==0 && year%100!=0) || (year%400==0) ))
then
    echo "Leap Year"
else
    echo "Not Leap Year"
fi
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter year:
2006
Not Leap Year
vaidehi@ubuntu:~$
```

- Write a shell script to compare two strings.

```bash
echo "Enter first string:"
read str1
echo "Enter second string:"
read str2

if [ "$str1" = "$str2" ]
then
    echo "Strings are equal"
else
    echo "Strings are not equal"
fi
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter first string:
Vaidehi
Enter second string:
Deshpande
Strings are not equal
vaidehi@ubuntu:~$
```

- Write a shell script to implement a menu-driven calculator using a case statement.

```bash
echo "1.Add 2.Subtract 3.Multiply 4.Divide"
echo "Enter choice:"
read ch

echo "Enter two numbers:"
read a b

case $ch in
1) echo "Result: $((a+b))" ;;
2) echo "Result: $((a-b))" ;;
3) echo "Result: $((a*b))" ;;
4) echo "Result: $((a/b))" ;;
*) echo "Invalid choice" ;;
esac
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
1.Add 2.Subtract 3.Multiply 4.Divide
Enter choice:
1
Enter two numbers:
8  9
Result: 17
vaidehi@ubuntu:~$ ./practical6.sh
1.Add 2.Subtract 3.Multiply 4.Divide
Enter choice:
4
Enter two numbers:
20 4
Result: 5
```

- Write a shell script for Employee Pay calculation.

```bash
echo "Enter basic salary:"
read basic

da=$((basic*40/100))
hra=$((basic*20/100))
gross=$((basic+da+hra))

echo "DA: $da"
echo "HRA: $hra"
echo "Gross Salary: $gross"
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter basic salary:
10000000
DA: 4000000
HRA: 2000000
Gross Salary: 16000000
vaidehi@ubuntu:~$
```

Conclusion:

This practical demonstrates how shell scripting can automate tasks like calculations, decision-making, and menu-driven programs. It helps in improving efficiency and reduces manual work in operating systems.

Error:

There was no such major error occurred.